LESSON PLAN · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

# Lesson 10: A Quiz Engine with Parallel Lists

Module 4 | 45 minutes | Data-driven programs | Grades 6–8 · ages 11–14

**Objective:** Store questions and answers in parallel lists and write one loop that runs a quiz of any length.

**Success criteria**

- I can store a table in parallel lists.
- I can write one engine loop for any dataset.
- I can explain data-driven design.

## Key vocabulary

- **parallel lists**: Lists whose items match up by position.
- **data-driven**: Behaviour controlled by data, not new code.
- **engine**: The general code that processes the data.
- **record**: One row of related data.

## You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

## Timed lesson plan (45 min)

| Time | Stage | What happens |
|---|---|---|
| 5 min | **Warm-up (unplugged)** | Card pairs (5 min): question cards and answer cards in two numbered piles; a quizmaster uses index numbers. |
| 5 min | **Watch** | Play the lesson video with captions; pause at each heading and ask a question. |
| 8 min | **Build along** | Build the quiz engine: learners follow the steps. |
| 14 min | **Perform** | Code Lab: quiz engine, Out of step |
| 9 min | **Probe** | Data vs code, activity |
| 4 min | **Wrap-up** | Exit ticket: What links item 3 of questions with item 3 of answers? |

## Unplugged warm-up

Card pairs (5 min): question cards and answer cards in two numbered piles; a quizmaster uses index numbers.

## Teaching the idea

Parallel lists model a table (records by index). A single traversal engine processes any dataset — separation of data and logic. Watch for mismatched list lengths.

**Analogy:** A board game: the rules (engine) stay the same; the question cards (data) change.

**Live demo:** Swap the quiz data to a new topic live without touching the engine.

## Questions to ask

| Ask | Listen for |
|---|---|
| How do the two lists connect? | By index. |
| How do you make a new quiz? | Change the lists. |
| What if the lists are different lengths? | Answers get out of step. |

## Differentiation

**Support**
Provide the filled lists and the loop skeleton.

**Stretch**
Shuffle the order using a random index and remove asked questions.

## Common misconceptions

| Misconception | How to fix it |
|---|---|
| Each question needs its own code | One loop handles them all. |
| Lists are linked automatically | Only by matching index. |

## Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

**Exit ticket**
What links item 3 of questions with item 3 of answers?

**Homework**
Write 5 question/answer pairs for a topic you're studying.

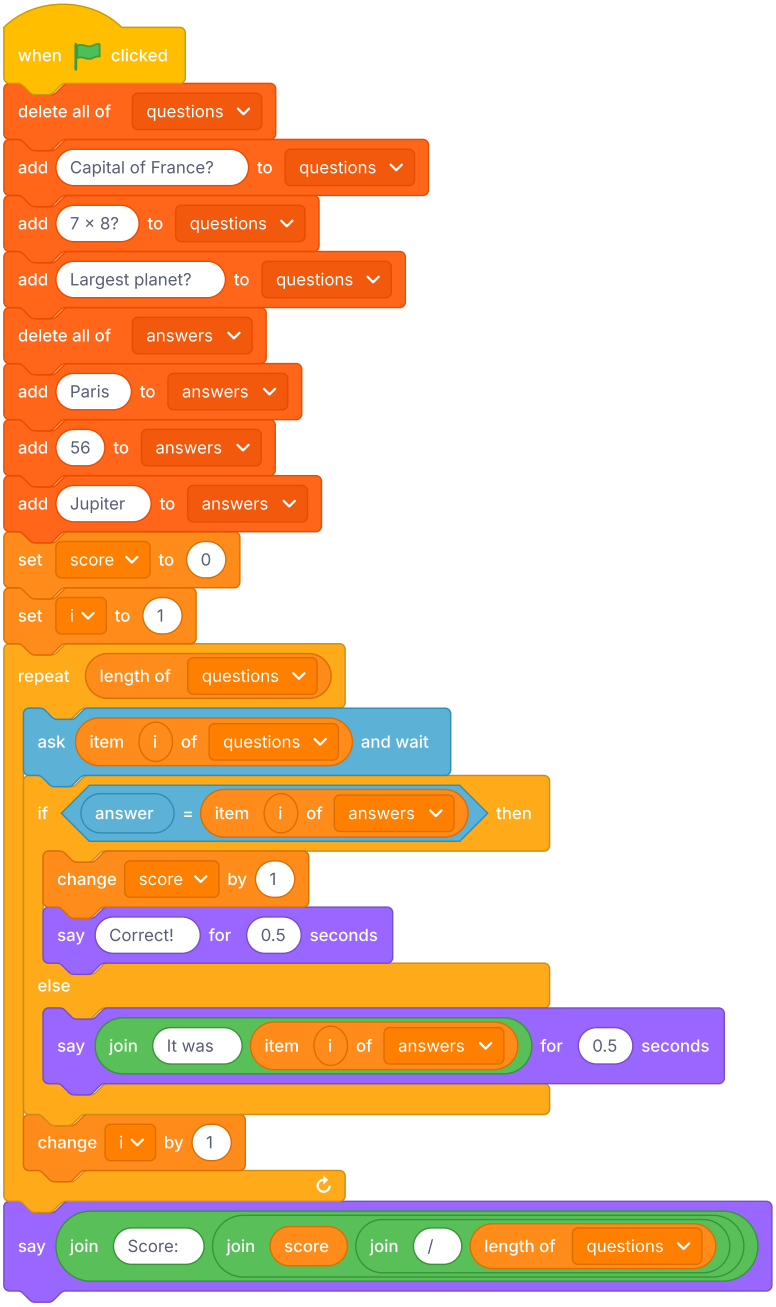
# Answer key — Kitabash activities

## 1. Code Lab: quiz engine (perform · code lab)

**Goals checked:** All correct → 3 · Wrong answers don't score · Asks from the list · Works for any length

**One solution:**

### Host




## 2. Out of step (perform · multiple choice)

**Answer:** No answer can match, so it's always marked wrong

**3. Data vs code** (probe · fill blank)

**Answers:** data · engine · data

**4. Explain your thinking** (probe · free response)

**Look for:** data, code, same. **Example:** A music app uses the same player code for every song; the data is the song files and the engine plays them. A flashcard app uses the same code for every deck; the data is the cards and the engine shows and scores them.

## Answer key — worksheet

1. **Key words:** parallel lists = Lists whose items match up by position. · data-driven = Behaviour controlled by data, not new code. · engine = The general code that processes the data. · record = One row of related data.
2. **Plan your code:** Goals: All correct → 3; Wrong answers don't score; Asks from the list; Works for any length.
3. **Out of step:** A. No answer can match, so it's always marked wrong — item 4 of answers is empty. Keep parallel lists the same length.
4. **Data vs code:** data · engine · data
5. **Explain your thinking:** A music app uses the same player code for every song; the data is the song files and the engine plays them. A flashcard app uses the same code for every deck; the data is the cards and the engine shows and scores them.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.